Computer Science & IT

Theory of Computation



CFL

Lecture No. 7



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Recap of Previous Lecture



- Topic
- Topic
- Topic

CFG

Topics to be Covered



- Topic
- Topic
- Topic

Context free grammars
+
String matching ✓

$L = \{w / n_a(w) = n_b(w)\} \rightarrow$ DCFL

$S \rightarrow aB / bA$

$A \rightarrow a / aS / bAA$

$B \rightarrow b / bS / aBB$

$S \rightarrow aSb / bSa / \epsilon$

abba ✗

$S \rightarrow aSb / bSa / SS / \epsilon$ ✓

$S \rightarrow aSbS / bSaS / \epsilon$ ✓

$S \rightarrow SaSb / SbSa / \epsilon$ ✓

Gate Which of the following are equivalent

a) $S \to aSb/\epsilon$ } $a^n b^n / n \geq 0$

b) $S \to aSb/ab/\epsilon$

c) $S \to aSb/ab$ } $a^n b^n / n \geq 1$

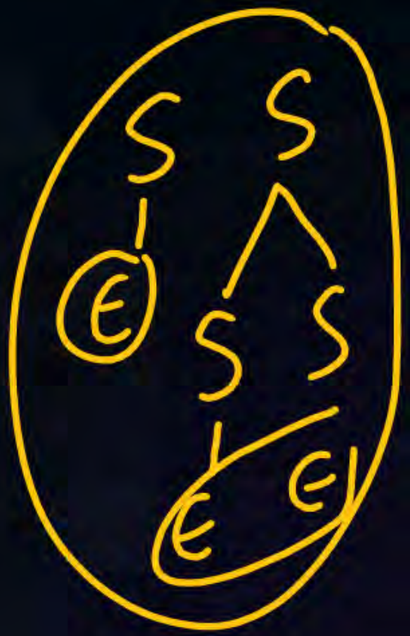
$L = \{ w / n_a(w) = 2 * n_b(w) \}$

aab, aba, baa

$S \rightarrow aSaSb / aSbSa / bSaSa / SS / \epsilon$

$S \rightarrow aSaSb / aSbSaS / bSasaS / \epsilon$

$S \rightarrow aSaSb / SaSbSa / SbSaSa / \epsilon$



$L = \{w / n_a(w) = 3n_b(w)\}$

aaaab aaaba, abaaa, baaa

$S \to aSaSaSb/aSaSbSa/aSbSaSa/bSaSaSa/SS/\epsilon$

$S \to aSaSaSbS/aSaSbSaS/aSbSaSaS/bSaSaSaS/\epsilon$

$S \to SaSaSaSb/SaSaSbSa/SaSbSaSa/SbSaSaSa/\epsilon$

Perfectly balanced parantheses

~~$L = \{w / n_(w) = n_)(w)\}$~~ ✗

$L = \{w / n_(w) = n_)(w)$ if v is any prefix of w, $n_(v) \geq n_)(v)\}$

((())) ()()

))((())() ()

$S \rightarrow (S) / SS / \epsilon$



((()))
))((

String matching ww^R / ww

Even length palindrome $ww^R / w \in (a,b)^*$ — CFL

Odd length palindrome $w(x)w^R / w \in (a,b)^*$ $x \in (a,b)$ — CFL

All palindromes $wxw^R / w \in (a,b)^*$ $x \in (\in, a, b)$ → CFL

Hash palindrome $w(\#)w^R / w \in (a,b)^*$ → DCFL

(# : special symbol)

$ww^R \cup (0,1)^* / w \in (0,1)^* \rightarrow (0,1)^*$



$ww^R / w \in (0,1)^*$

✗ Reg ✗ DCFL ✓ CFL

✓ NPDA

$ww^R / w \in a^*$

$w - \epsilon$

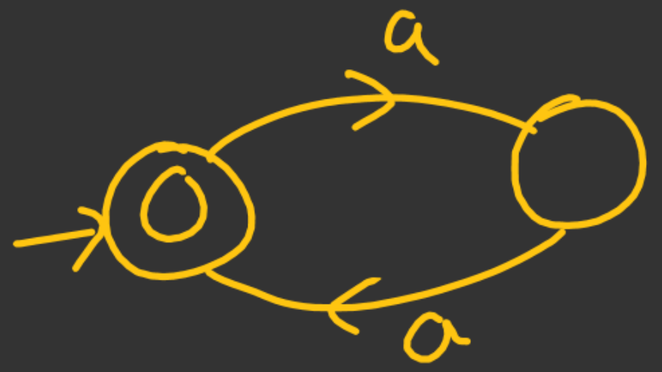
$ww^R - \epsilon$

$w = a$

ww^R a a ✓

$w = aa$ ✓

ww^R $\underline{aa(aa)}$ $aaa(aaa)$ $(aa)^*$



$a^0, aa, aaaa, a^6, a^8 \ldots$

$w(w^R)^{*^0}$ / $w \in a^*$

$w \cdot \{\epsilon, w^R, (w^R)^2, (w^R)^3 \ldots\}$

$\{w, ww^R, w(w^R)^2, w(w^R)^3, \ldots\}$

a^* ∪ anything $= a^*$



universal

Reg

$\{w\#w^R / w \in a^*\}$



$\#$ $a\#a$ $aa\#aa$ $aaa\#aaa$

$a^n \# a^n / n \geq 0$

PDA DCFL

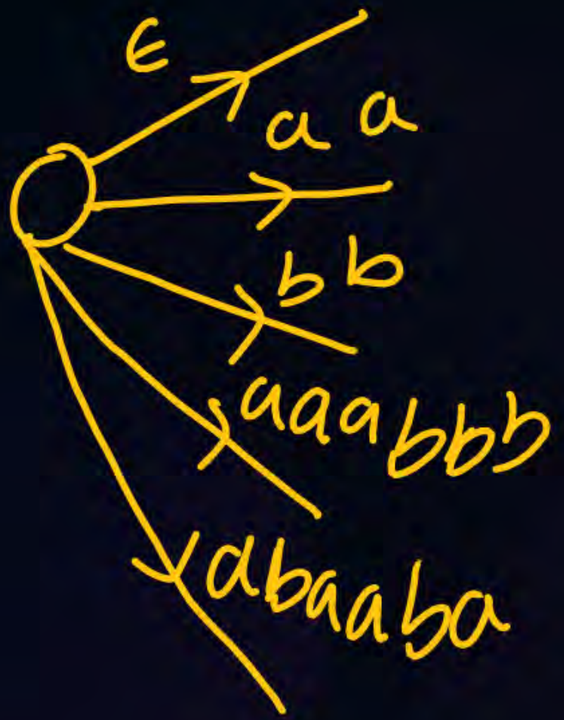
$ww^R / w \in \{a,b\}^*$ and $|w| \leq 10$

↓

finite

↓

Reg



Every finite langage is Reg

$w \; ww / w \in (\tilde{a}, \tilde{b})$

$(a+b)(a+b)(a+b)$

~~strings~~ → Finite → Regular

$\frac{aaa}{bbb}$

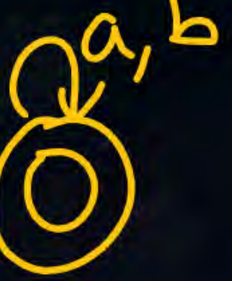
$w(w^R)^* / w \in (a,b)^*$

$w \cdot \{\epsilon, w^R, (w^R)^2, (w^R)^3 \ldots\ldots\}$

$w, \underbrace{ww^R, w(w^R)^2}$

$\rightarrow (a+b)^*$

Reg



$(a+b)^* \rightarrow$

Una

$w^* w^R / w \in (a,b)^*$

$\{\in, w, w^2, w^3, \ldots\}.w^R$

$w^R, (ww^R, w^2w^R, w^3w^R \ldots)$

$(a+b)^* \rightarrow$ Reg

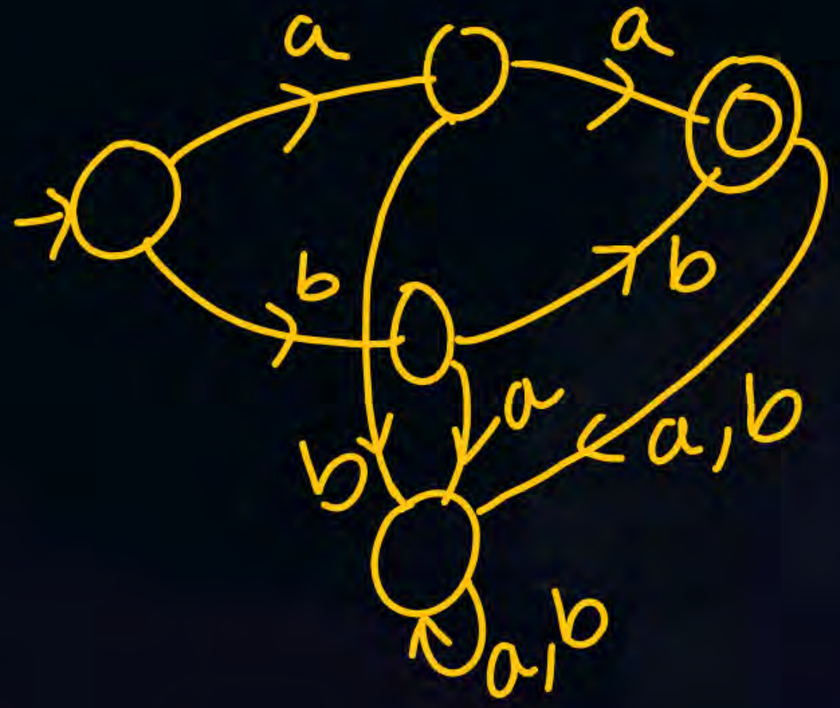
ww^R



$WW^R / W \in \{a,b\}$ How many stats in min DFA

$w = a \quad WW^R = aa$ ✓

$w = b \quad WW^R = bb$ ✓



$w(w^R)^* / w \in (a+b)^+$

$w(\in, w^R, (w^R)^2, (w^R)^3 \ldots)$





$w^* w^R / w \in (a,b)$

$w = a$

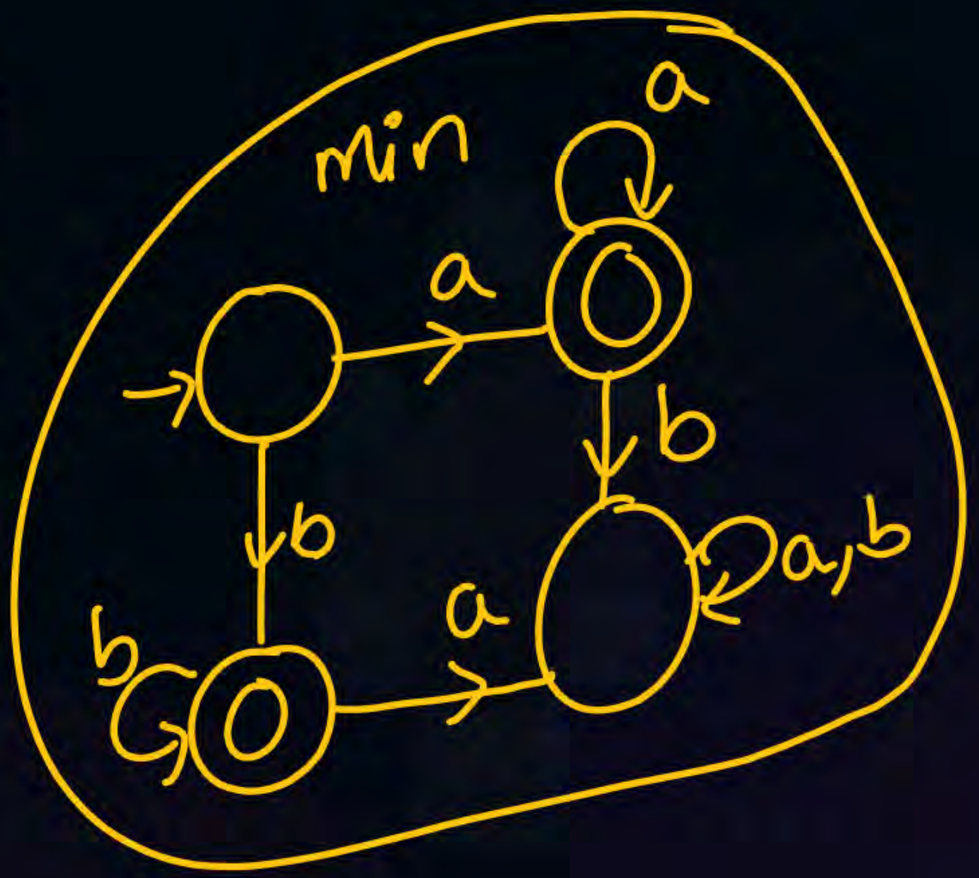
$w^* w^R = a^* a$

$w = b$

$w^* w^R = b^* b$

$a^* a + b^* b = a a^* + b b^*$

$a^+ + b^+$



$w^* w^R / w \in (a,b)^*$

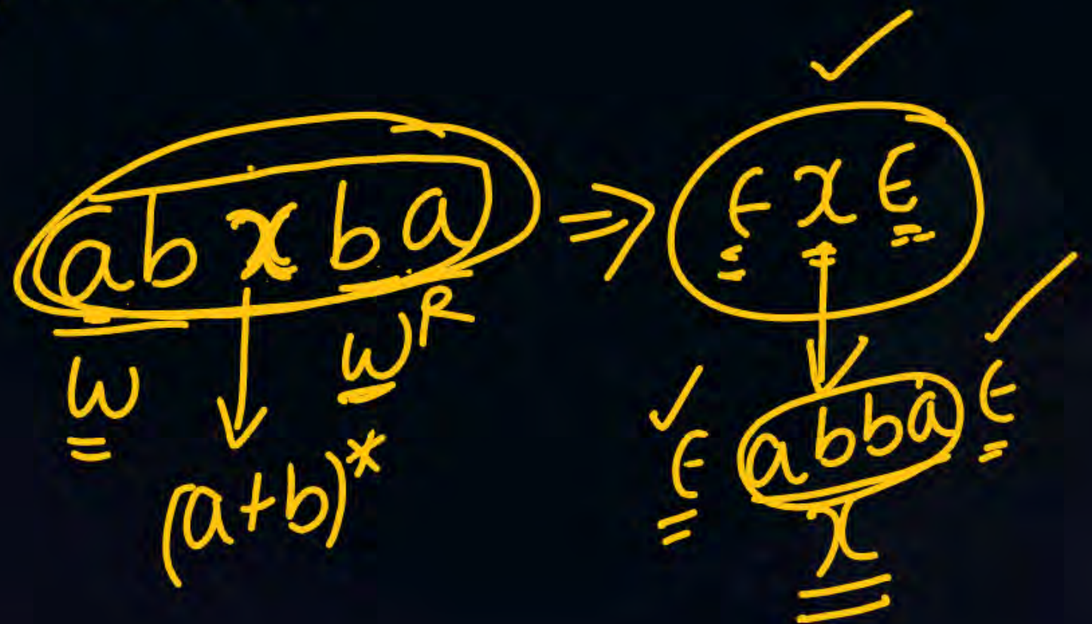
$\in$ w^R

$(a+b)^* \rightarrow$ Reg

$wxw^R / w \in (a,b)^*, \; x \in (a,b)^*$

$\epsilon \mid \epsilon$

$(a+b)^* \rightarrow reg$



$wxw^R / w \in (a,b)^* \quad x \in (a,b)^+$

$\epsilon \quad | \quad \epsilon$

$(a+b)^+ \rightarrow$ Reg

$\epsilon \notin L$

$(wxw^R / \ w \in \{a,b\}^* \quad x \in \{a,b\})$

DCFL ✗

CFL ✓

odd length palindrome

$w x w^R / w \in (a,b)^*, x \in (a,b)^*$

$a(a+b)^* a$
$b(a+b)^* b$ ✓

$\underline{w}xw^R / w \in (a,b)^*$ $\quad x \in (a,b)^*$

$a x a$
$b x b$

$a(a+b)a = aaa, aba$
$b(a+b)b = bab, bbb$

xww^Rx ww^Rx $w=ab$ wxw^R

x

$ab(a+b)^*b$ a $w=ba$

$L_1 = wxw^R / w\in(a,b)^+, x\in(a,b)^*$

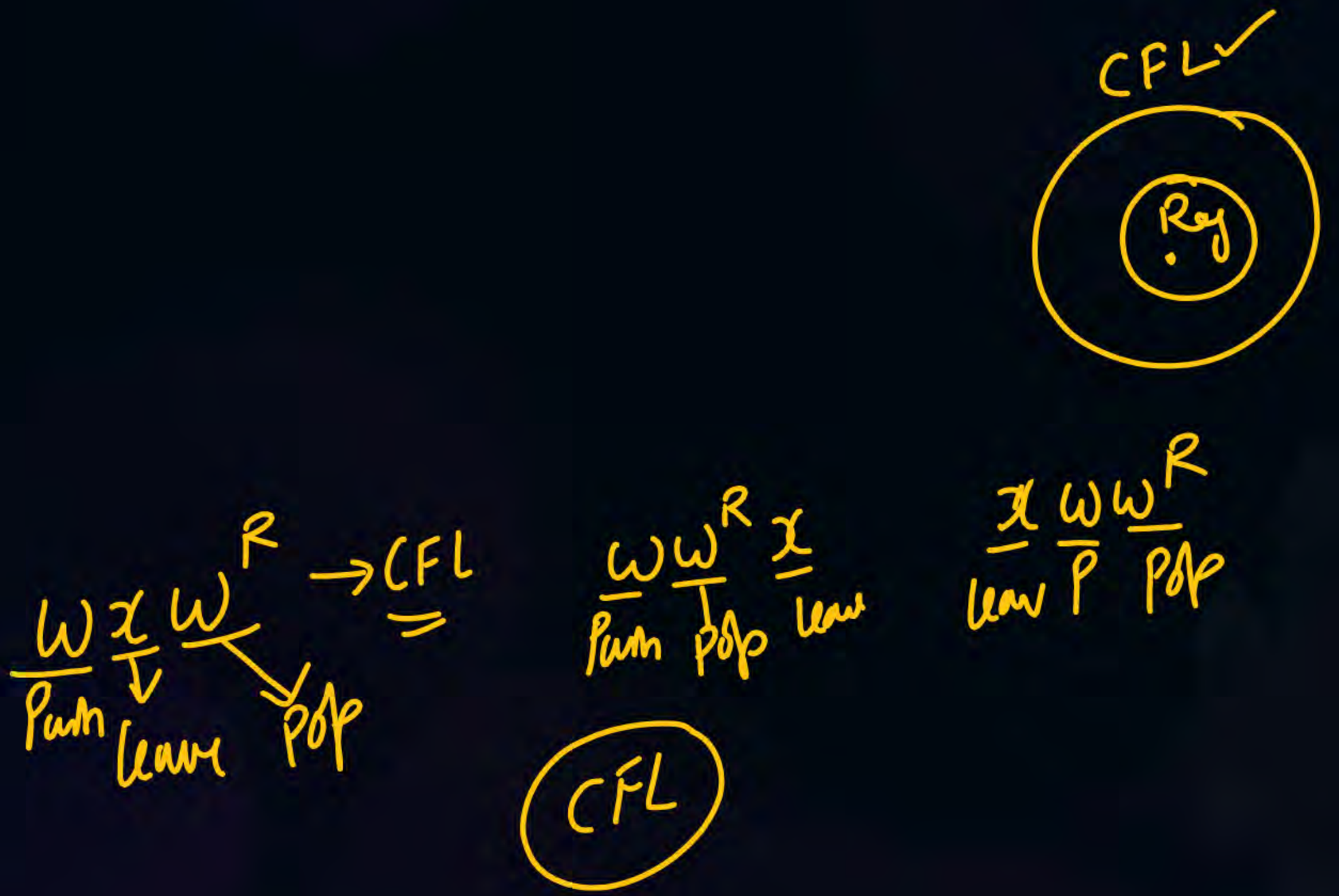
$a(a+b)^*a + b(a+b)^*b$ → Reg

L_2 L_3

$L_1 = L_2 + L_3$ $L_1 \supseteq L_2 + L_3$

x

$b(a(a+b)^*a)b$


CFL
Reg
$w x w^R \rightarrow$ CFL
Push
Leave
Pop
$w w^R x$
Push
Pop
Leave
CFL
$x w w^R$
Leave
Push
Pop

xww^R – ✗CFL

$L_1 = \{xww^R / x, w \in (a,b)^+\}$

$w = ab$

$(a+b)^+ \; ab \; ba$ (w, w^R)

$L_2 = (a+b)^+ aa$ ✓

$L_3 = (a+b)^+ bb$ ✓

$L_1 \overset{?}{=} L_2 + L_3$

$L_1 \supseteq L_2 + L_3$ ✓

✗ Reg

✓ CFL

✗ DCFL

$a\,(b(a+b)^+b)\,a$ — w | $(a+b)^+$ | w^R

$b\,(a(a+b)^+a)\,b$ — w | w^R

$L_1 = \{wxw^R \mid x, w \in (a,b)^+\}$

$L_2 = a(a+b)^+a$

$L_3 = b(a+b)^+b$

$L_1 = L_2 + L_3$

Reg

$w = ab$

$a\,(b(a+b)^+b)\,a$ — w | w^R

$a\,(bb(a+b)^+bb)\,a$ — w | x | w^R

$L_1 = \overline{w w^R x} \;/\; w \in (a+b)^+$

$L_2 = aa(a+b)^+$

$L_3 = bb(a+b)^+$

CFL ✓

Regx

DCFLX

$L_1 \overset{?}{=} L_2 + L_3$ ✓

$w = ab$



Derivation :-

The process of deriving a string

$S \to AB$

$A \to a$

$B \to b$

$S \Rightarrow \underline{AB} \Rightarrow aB \Rightarrow ab$ LMD

$S \Rightarrow AB \Rightarrow Ab \Rightarrow ab$ RMD

G

$S \to AB$
$A \to aA/\epsilon$
$B \to bB/\epsilon$

$\tilde{\omega} \in L(G)$

$1 \le DT, 1 \le LMD, 1 \le RMD$

LMD

$S \Rightarrow AB \Rightarrow aAB \Rightarrow aaAB \Rightarrow aaaB \Rightarrow aaabB \Rightarrow aaab$

$S \Rightarrow AB \Rightarrow AbB \Rightarrow Ab \Rightarrow aAb \Rightarrow aaAb \Rightarrow aaaAb \Rightarrow aaab$



For every derivation tree, there is only one LMD
and only one RMD

If $w \in L$, there will be atleast one derivation tree, atleast one LMD
and atleast one RMD

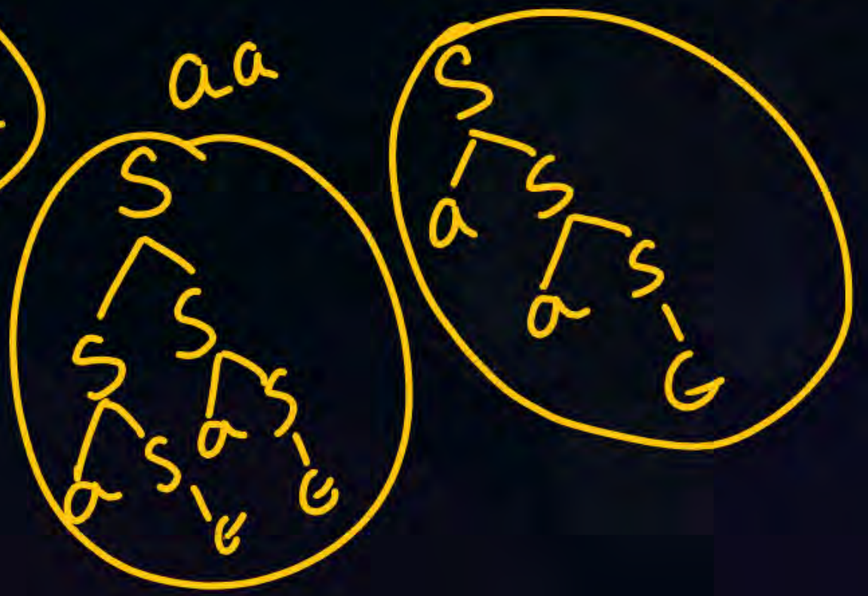
A grammar is unambiguous if it has exactly one derivation tree and exactly one LMD and exactly one RMD for all strings in the language

A grammar is ambiguous $\exists w \in L$, such that w has more than one derivation tree. So there will be more than 1 LMD and more than 1 RMD

$S \to aS/bS/SS/\epsilon$

amb grammar

aa



In an ambiguous grammar, we may have more than one derivation tree for a string

In unambiguous grammar, every string has exactly 1 Derivation tree

exactly 1 LMD

exactly 1 RMD

If 'w' has 1 derivation tree $\Leftrightarrow$ w has 1 RMD

$\Leftrightarrow$ w has 1 LMD

If 'w' has atleast 1 DT $\Leftrightarrow$ w has atleast 1 RMD

$\Leftrightarrow$ w has atle 1 LMD

5 min Break

$S \to Xa / Ya$

~~$X \to Za$~~

$Z \to Sa / \epsilon$

~~$Y \to Wa$~~

$W \to Sa$

$S \to Zaa / Waa$

$S \to Saaa / aa / Sa\epsilon$

$S \to Saaa / aa$

What is the minimum no of states in DFA for this language?

a) 2 b) 3 c) 4 d) 5

$aa(aaa)^*$

No of a's $\cong 2 \pmod 3$

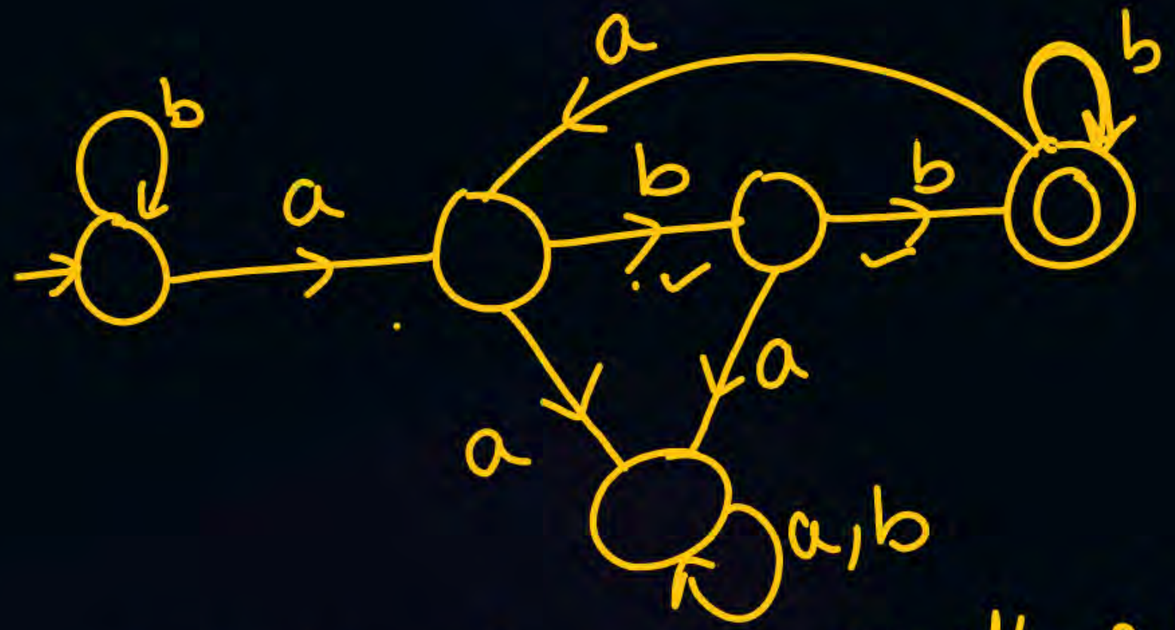
A language L satisfies the pumping lemma for regular languages and PL for CFL. which of following is true

a) L is necessarily Regular

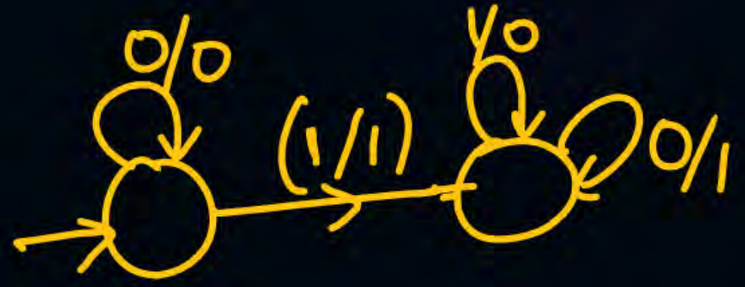
b) L is necessarily CFL but not Regular

c) L is necessarily non Regular

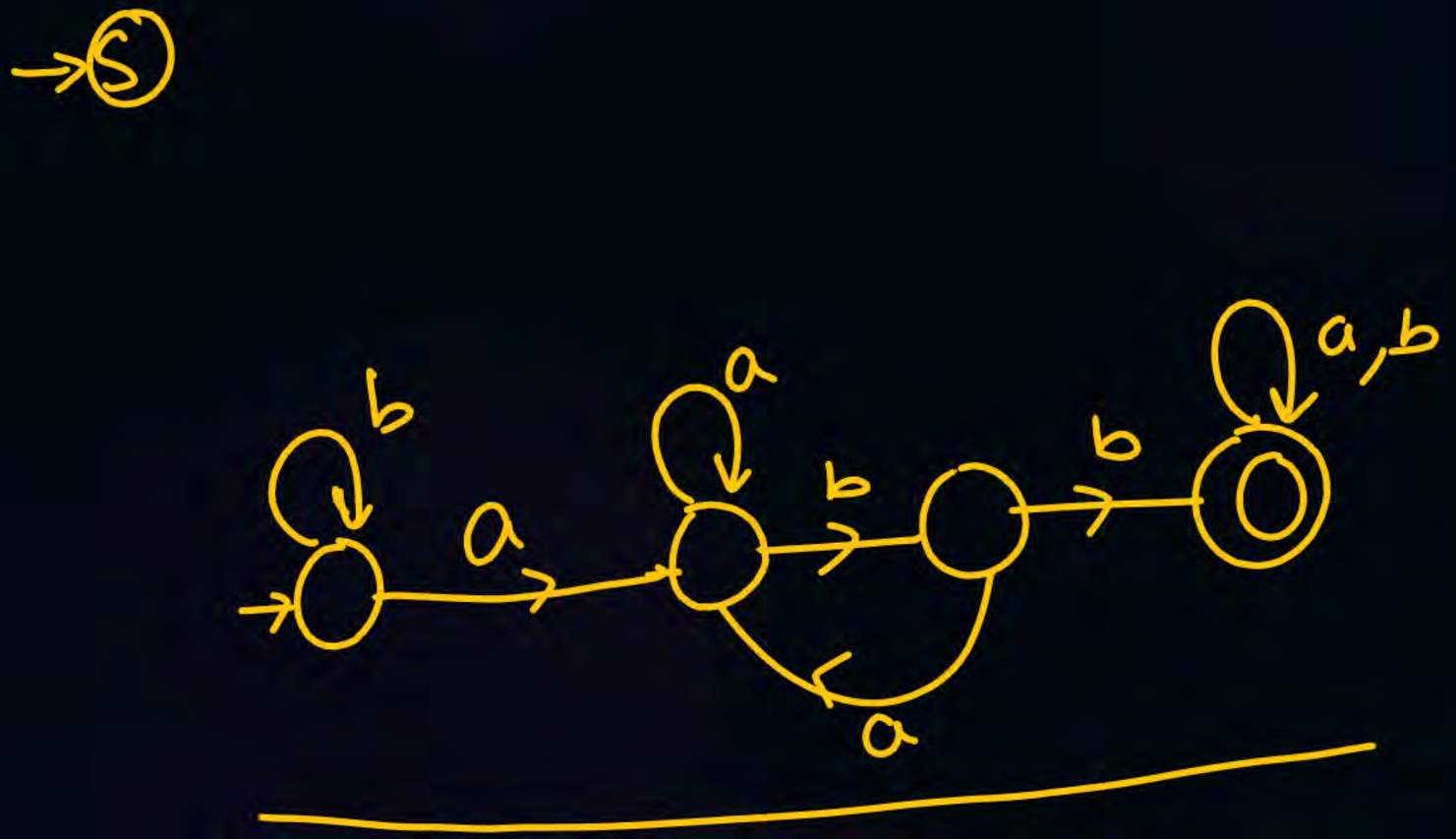
d) None of the above

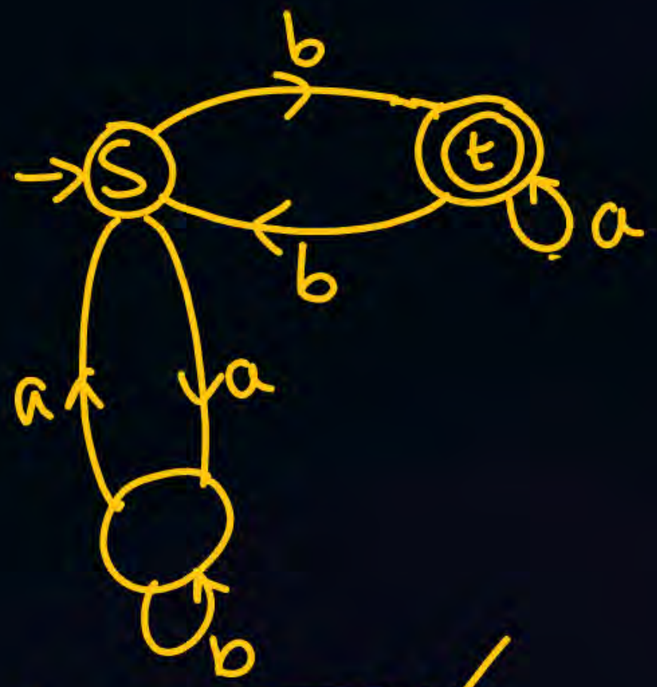


a) a is followed by exactly 2 b's
b) a is " " " atleast 2 b's
c) w abb is a substring
d) does not contain 'aa'



a) It Computes 1's Complement
b) It Computes 2's Complement ✓
c) It increments the i/p number
d) It decrements i/p number


S
b
a
a
b
b
a,b
a



u = abbabba

v = bab

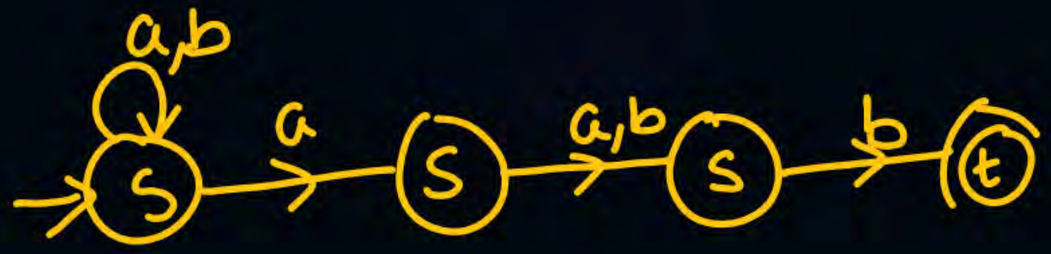
w = aabb

a) It accepts u and v but not w

b) It accepts u, v, w

c) It rejects u v w

d) accepts u but rejects v and w



a) $(a+b)^* a(a+b)b$

b) $(ab)^*$

c) $(a+b)^* a(a+b)^* b(a+b)^*$

d) $(a+b)^*$

Which of the following is not true

a) Every language has a regular superset

b) Every language has a regular subset

X c) Every subset of regular language is reg

d) Every subset of finite language is regular

$L \subseteq (a+b)^*$

$\phi \subseteq L$

$\underset{NR}{a^n b^n} \subseteq \underset{R}{(a+b)^*}$

Finite $\subset$ Finite $\rightarrow$ Reg

THANK - YOU